Patient and mutation filtering analysis for actionable mutation modeling

Overview

This document presents data to inform decisions on filtering and aggregating clinically actionable mutations. The data are primarily aimed at aggregations at the level of individual patients. The results are organized into three sections

1. Part 1 shows results for LEVEL 2 actionable events as assigned by OncoKB. The most common mutations are described at the level of cancer type, gene, and amino acid change.
2. Part 2 shows results related to mutation co-occurrence at the level of Level 1 and Level 2 events
3. Part 3 shows data on read coverage data and how filtering on this metric could affect patient-level aggregation.

Results are restricted to a subset of cancer types that have 250 or more patients with actionable events.

Loading data

Aggregation of data by cancer type

Data split by OncoTree codes

```
knitr::kable(outCntTbl, )
```

OT code	cancer type	cancer type detailed	number of patients	actionable events
LUAD	Lung Cancer	Lung Adenocarcinoma	23678	10286 (43.4%)
IDC	Breast Cancer	Breast Invasive Ductal Carcinoma	10117	4450 (44%)
BRCA	Breast Cancer	Invasive Breast Carcinoma	4050	1956 (48.3%)
MEL	Melanoma	Melanoma	4832	1556 (32.2%)

OT code	cancer type	cancer type detailed	number of patients	actionable events
GIST	Gastrointestinal Stromal Tumor	Gastrointestinal Stromal Tumor	1896	1511 (79.7%)
COAD	Colorectal Cancer	Colon Adenocarcinoma	11806	1276 (10.8%)
THPA	Thyroid Cancer	Papillary Thyroid Cancer	1600	1070 (66.9%)
ILC	Breast Cancer	Breast Invasive Lobular Carcinoma	1635	1016 (62.1%)
PRAD	Prostate Cancer	Prostate Adenocarcinoma	7026	929 (13.2%)
AML	Acute myeloid leukemia	Acute myeloid leukemia	2825	868 (30.7%)
BLCA	Bladder Cancer	Bladder Urothelial Carcinoma	4498	777 (17.3%)
SKCM	Melanoma	Cutaneous Melanoma	2118	714 (33.7%)
NSCLC	Non-Small Cell Lung Cancer	Non-Small Cell Lung Cancer	2366	689 (29.1%)
ODG	Glioma	Oligodendroglioma	682	651 (95.5%)
HGSOC	Ovarian Cancer	High-Grade Serous Ovarian Cancer	4725	603 (12.8%)
ASTR	Glioma	Astrocytoma	821	559 (68.1%)
MDLC	Breast Cancer	Breast Mixed Ductal and Lobular Carcinoma	570	313 (54.9%)
BRCNOS	Breast Cancer	Breast Invasive Carcinoma, NOS	595	289 (48.6%)
BRCANOS	Breast Cancer	Breast Invasive Cancer, NOS	610	287 (47%)
COADREAD	Colorectal Cancer	Colorectal Adenocarcinoma	4145	283 (6.8%)
IHCH	Hepatobiliary Cancer	Intrahepatic Cholangiocarcinoma	1210	281 (23.2%)

Most common mutated genes for each cancer type

top five actionable genes for each cancer type

Top 20 most common actionable events by gene

```
knitr::kable(outGeneTbl[1:20,], )
```

cancer type detailed	gene	count and perc of actionable events
Lung Adenocarcinoma	EGFR	5596 (54.4%)
Breast Invasive Ductal Carcinoma	PIK3CA	3259 (73.2%)
Lung Adenocarcinoma	KRAS	3129 (30.4%)
Melanoma	BRAF	1556 (100%)
Gastrointestinal Stromal Tumor	KIT	1462 (96.8%)
Colon Adenocarcinoma	BRAF	1276 (100%)
Invasive Breast Carcinoma	PIK3CA	1250 (63.9%)
Papillary Thyroid Cancer	BRAF	1070 (100%)
Breast Invasive Lobular Carcinoma	PIK3CA	775 (76.3%)

cancer type detailed	gene	count and perc of actionable events
Bladder Urothelial Carcinoma	FGFR3	771 (99.2%)
Cutaneous Melanoma	BRAF	714 (100%)
Lung Adenocarcinoma	MET	623 (6.1%)
Oligodendroglioma	IDH1	582 (89.4%)
Lung Adenocarcinoma	ERBB2	544 (5.3%)
Astrocytoma	IDH1	516 (92.3%)
Breast Invasive Ductal Carcinoma	PTEN	477 (10.7%)
High-Grade Serous Ovarian Cancer	BRCA1	397 (65.8%)
Lung Adenocarcinoma	BRAF	394 (3.8%)
Breast Invasive Ductal Carcinoma	AKT1	390 (8.8%)
Acute myeloid leukemia	FLT3	342 (39.4%)

Most common coding mutations for each cancer type

Ten most common mutations for each cancer type

```
#knitr::kable(outMutTbl, )
```

Most common Level 2 events by cancer type

```
knitr::kable(outMutTbl[1:15,], )
```

cancer type detailed	count and perc of actionable events
Breast Invasive Lobular Carcinoma	88 (8.7%)
Cutaneous Melanoma	46 (6.4%)
Bladder Urothelial Carcinoma	48 (6.2%)
Melanoma	96 (6.2%)
Breast Mixed Ductal and Lobular Carcinoma	19 (6.1%)
Colon Adenocarcinoma	69 (5.4%)
Colorectal Adenocarcinoma	14 (4.9%)
Invasive Breast Carcinoma	78 (4%)
Breast Invasive Cancer, NOS	10 (3.5%)
Breast Invasive Ductal Carcinoma	144 (3.2%)
Intrahepatic Cholangiocarcinoma	8 (2.8%)
Breast Invasive Carcinoma, NOS	8 (2.8%)
Non-Small Cell Lung Cancer	18 (2.6%)
Gastrointestinal Stromal Tumor	37 (2.4%)
Lung Adenocarcinoma	89 (0.9%)

Most common Level 2 events for each cancer type & gene

```
knitr::kable(outMutTbl[1:15,], )
```

cancer type detailed	gene	count and perc of actionable events
Breast Invasive Lobular Carcinoma	ERBB2	86 (8.5%)
Cutaneous Melanoma	KIT	46 (6.4%)
Bladder Urothelial Carcinoma	FGFR3	48 (6.2%)
Melanoma	KIT	96 (6.2%)
Colon Adenocarcinoma	POLE	66 (5.2%)
Breast Mixed Ductal and Lobular Carcinoma	ERBB2	16 (5.1%)
Colorectal Adenocarcinoma	POLE	14 (4.9%)
Invasive Breast Carcinoma	ERBB2	71 (3.6%)
Intrahepatic Cholangiocarcinoma	KRAS	8 (2.8%)
Gastrointestinal Stromal Tumor	PDGFRA	37 (2.4%)
Breast Invasive Ductal Carcinoma	ERBB2	104 (2.3%)
Breast Invasive Cancer, NOS	ERBB2	6 (2.1%)
Breast Invasive Carcinoma, NOS	ERBB2	5 (1.7%)
Breast Invasive Cancer, NOS	AKT1	3 (1%)
Breast Invasive Carcinoma, NOS	AKT1	3 (1%)

Most common Level 2 events for each cancer type, gene, and HGVSp entry

```
knitr::kable(outMutTbl[1:15,], )
```

cancer type detailed	gene	protein mutation	count and perc of actionable events
Melanoma	KIT	p.L576P	40 (2.6%)
Breast Invasive Lobular Carcinoma	ERBB2	p.L755S	32 (3.1%)
Colon Adenocarcinoma	POLE	p.P286R	26 (2%)
Invasive Breast Carcinoma	ERBB2	p.L755S	23 (1.2%)
Lung Adenocarcinoma	ALK	p.G1202R	21 (0.2%)
Breast Invasive Ductal Carcinoma	ERBB2	p.G778_P780dup	20 (0.4%)
Lung Adenocarcinoma	ALK	p.L1196M	19 (0.2%)
Gastrointestinal Stromal Tumor	PDGFRA	p.V561D	19 (1.3%)
Cutaneous Melanoma	KIT	p.L576P	18 (2.5%)
Breast Invasive Ductal Carcinoma	ERBB2	p.L755S	17 (0.4%)
Breast Invasive Ductal Carcinoma	ERBB2	p.V777L	17 (0.4%)
Colon Adenocarcinoma	POLE	p.V411L	16 (1.3%)
Invasive Breast Carcinoma	ERBB2	p.V777L	15 (0.8%)
Melanoma	KIT	p.K642E	15 (1%)
Cutaneous Melanoma	KIT	p.K642E	14 (2%)

Look at actionable mutation co-occurrence

```
perOncoKBEvntCnts <- svCompiled %>%
  dplyr::rename(SequenceCenter = Center) %>%
  #dplyr::left_join(assayInfo,by=c("SEQ_ASSAY_ID_mod"="SEQ_ASSAY_ID")) %>%
  dplyr::mutate(isOncoKbLevel1=ONCOKB_HIGHEST_LEVEL_SUMMARY=="LEVEL_1" & !is.na(ONCOKB_HIGHEST_LEVEL_
                isOncoKbLevel2=ONCOKB_HIGHEST_LEVEL_SUMMARY=="LEVEL_2" & !is.na(ONCOKB_HIGHEST_LEVEL_
                isOncoKbLevel3=ONCOKB_HIGHEST_LEVEL_SUMMARY=="LEVEL_3" & !is.na(ONCOKB_HIGHEST_LEVEL_
                isOncoKbLevelAny=ONCOKB_HIGHEST_LEVEL_SUMMARY=="LEVEL_" & !is.na(ONCOKB_HIGHEST_LEVEL
                isOncoKbLevel2or1=(ONCOKB_HIGHEST_LEVEL_SUMMARY=="LEVEL_1" | ONCOKB_HIGHEST_LEVEL_SUM
  dplyr::group_by(Tumor_Sample_Barcode) %>%
  dplyr::summarise(nLevel1Events=sum(isOncoKbLevel1),
                   nLevel2Events=sum(isOncoKbLevel2),
                   nLevelAnyEvent=sum(isOncoKbLevelAny))
  #dplyr::arrange(desc(isOncoKbLevel1),desc(isOncoKbLevel2or1),desc(isOncoKbLevelAny)) %>%
  #dplyr::filter(dplyr::row_number()==1) %>%

oncoKBEventCnts <- perOncoKBEvntCnts %>%
  dplyr::group_by(nLevel1Events,nLevel2Events,nLevelAnyEvent) %>%
  dplyr::summarise(n_patients=n(),
                   percent_total=100*n_patients/dim(perOncoKBEvntCnts)[1],
                   event_counts = paste0(n_patients, " (", round(percent_total, 1), "%)")) %>%
  dplyr::arrange(desc(n_patients))
```

```
`summarise()` has grouped output by 'nLevel1Events', 'nLevel2Events'. You can
override using the `.groups` argument.
```

```
knitr::kable(oncoKBEventCnts[1:15,], )
```

nLevel1Events	nLevel2Events	nLevelAnyEvent	n_patients	percent_total	event_counts
0	0	0	175777	82.8406076	175777 (82.8%)
1	0	0	31743	14.9599174	31743 (15%)
2	0	0	2256	1.0632131	2256 (1.1%)
0	1	0	1753	0.8261581	1753 (0.8%)
1	1	0	263	0.1239473	263 (0.1%)
3	0	0	227	0.1069811	227 (0.1%)
0	2	0	75	0.0353462	75 (0%)
4	0	0	29	0.0136672	29 (0%)
2	1	0	28	0.0131959	28 (0%)
1	2	0	12	0.0056554	12 (0%)
5	0	0	9	0.0042415	9 (0%)
6	0	0	4	0.0018851	4 (0%)
2	2	0	3	0.0014138	3 (0%)
3	1	0	2	0.0009426	2 (0%)
0	3	0	1	0.0004713	1 (0%)